

Edexcel A-Level Chemistry — Core Practicals

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CP1: Measure the Molar Volume of a Gas

Background Theory

The **molar volume** of a gas is the volume occupied by one mole of any gas at a stated temperature and pressure (about $24.0 \text{ dm}^3 \text{ mol}^{-1}$ at room temperature and pressure). In this practical, weighed samples of calcium carbonate react with an **excess** of ethanoic acid and the carbon dioxide is collected and measured. Because each mole of CaCO_3 gives one mole of CO_2 , the volume of gas per mole of carbonate gives the molar volume.



Ethanoic acid (a weak acid) is used rather than a strong acid so that the reaction is slower, giving time to replace the bung before much gas escapes.

Apparatus

- Boiling tube with bung and delivery tube
- 100 cm^3 measuring cylinder, inverted and full of water in a trough (or a 100 cm^3 gas syringe)
- About 30 cm^3 of 1 mol dm^{-3} ethanoic acid, powdered calcium carbonate
- Small test tube for weighing, balance reading to 0.01 g or better, clamp stand

Method

1. Put 30 cm^3 of ethanoic acid in the boiling tube and connect the delivery tube to the inverted, water-filled measuring cylinder.
2. Weigh a small test tube containing about 0.05 g of calcium carbonate.
3. Remove the bung, tip the carbonate into the acid, and **replace the bung immediately**.
4. When fizzing stops, read the volume of gas collected.
5. Reweigh the empty test tube; the mass of CaCO_3 used is the difference.
6. Repeat with fresh acid, increasing the mass of carbonate by about 0.05 g each time, up to about 0.40 g (larger masses would give more gas than the cylinder holds).

Analysis

- Calculate moles of $\text{CaCO}_3 = \text{mass} / 100.1$ for each run.

- Plot **volume of CO₂ (y-axis)** against **moles of CaCO₃ (x-axis)** and draw a line of best fit.
- The **gradient** is the molar volume (convert cm³ mol⁻¹ to dm³ mol⁻¹ by dividing by 1000).
- If the line does not pass through the origin, a constant amount of gas has been lost (or gained) in every run.

Safety

- Ethanoic acid is an irritant at this concentration — wear eye protection.
- Check the delivery tube is not blocked so that pressure cannot build up.

Error Analysis

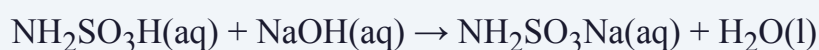
Source of Error	Type	Improvement
Gas lost before the bung is replaced	Systematic (volume too low)	Hang the carbonate in a small tube inside the flask and tip it over after sealing
CO ₂ dissolves slightly in the water in the trough and in the acid	Systematic (volume too low)	Use a gas syringe instead of collecting over water
Low precision of the balance for small masses	Random (large % uncertainty)	Use a 3 d.p. balance; weigh by difference
Reading the measuring cylinder scale	Random	Read at eye level; use a gas syringe with finer graduations

Exam Tip: Using a graph of volume against moles from several runs is better than a single measurement: it averages out random error, and a non-zero intercept reveals a systematic loss. The acid must be in **excess** so that the carbonate is the limiting reagent in every run. Be ready to calculate percentage uncertainties for the mass and the volume and to combine them.

CP2: Prepare a Standard Solution from a Solid Acid and Use It to Find the Concentration of Sodium Hydroxide

Background Theory

A **standard solution** has an accurately known concentration. It is made by dissolving an accurately weighed mass of a pure, stable solid in water and making the solution up to an exact volume in a volumetric flask. A solid acid such as **sulfamic acid** ($\text{NH}_2\text{SO}_3\text{H}$, a monoprotic acid, $M = 97.1 \text{ g mol}^{-1}$) is suitable because it is solid, stable in air and not hygroscopic. Sodium hydroxide cannot itself be used to make a standard solution because the solid absorbs water and carbon dioxide from the air, so its concentration must be found by titration against a standard acid.



Apparatus

- Weighing bottle, balance (2 or 3 d.p.), 250 cm^3 beaker, glass rod, funnel, wash bottle of distilled water
- **250 cm^3 volumetric flask** with stopper
- 25.0 cm^3 volumetric pipette and filler, 50 cm^3 burette, conical flasks, white tile
- Phenolphthalein or methyl orange indicator

Method: Making the Standard Solution

1. Weigh the weighing bottle containing about 2.5 g of sulfamic acid. Tip the solid into a beaker and reweigh the bottle; the mass used is the difference.
2. Dissolve the solid in about 100 cm^3 of distilled water, stirring with a glass rod.
3. Transfer the solution through a funnel into the volumetric flask. Rinse the beaker, rod and funnel with distilled water and add all rinsings to the flask.
4. Make up to the mark with distilled water, adding the last few drops with a dropping pipette so the **bottom of the meniscus** sits on the line at eye level.
5. Stopper and invert the flask several times to mix.

Method: Titration

1. Rinse the pipette with sodium hydroxide solution, then pipette 25.0 cm^3 into a conical flask and add a few drops of indicator.
2. Rinse the burette with the sulfamic acid solution, fill it, and make sure the jet is full with no air bubbles.
3. Do a rough titration, then accurate titrations adding the acid dropwise near the end-point while swirling.
4. Repeat until you have **concordant titres** (within 0.10 cm^3) and calculate the mean.

Key Practical Points

- Burette readings to **2 decimal places** (nearest 0.05 cm^3)
- **Rinse** the burette with the solution it will contain and the pipette with the solution it will deliver; rinse the conical flask with distilled water only
- Use a **white tile** to see colour changes clearly
- Indicator: phenolphthalein (colourless in acid, pink in alkali) or methyl orange (red in acid, yellow in alkali) both work for a strong acid with a strong base; the specification expects you to use both

Calculation

Concentration of acid = $(\text{mass} / 97.1) / 0.250 \text{ mol dm}^{-3}$

Moles of acid in mean titre = concentration x (titre / 1000)

Moles of NaOH = moles of acid (1 : 1 ratio)

[NaOH] = moles of NaOH / 0.0250 dm^3

Safety

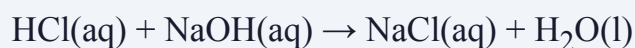
- Sodium hydroxide solution is corrosive to eyes; sulfamic acid is an irritant. Wear eye protection.
- Use a pipette filler; never pipette by mouth.

Exam Tip: A common question asks why a wet conical flask does NOT cause an error: the extra water does not change the moles of alkali already in the flask, so the same volume of acid is still needed. But a wet pipette or wet burette WOULD cause an error because the water dilutes the solution, changing its effective concentration. You may also be asked why NaOH is not used as a primary standard (it absorbs water and CO₂ from the air).

CP3: Find the Concentration of a Solution of Hydrochloric Acid

Background Theory

This titration builds on CP2 and focuses on **measurement uncertainty**. The hydrochloric acid is too concentrated to titrate directly, so it is first **diluted accurately** (for example, 25.0 cm³ made up to 250 cm³, a tenfold dilution). The diluted acid is then titrated against sodium hydroxide of known concentration.



Apparatus

- 25.0 cm³ volumetric pipette and filler, 250 cm³ volumetric flask
- 50 cm³ burette, conical flasks, white tile, indicator
- Sodium hydroxide solution of known concentration (e.g. about 0.1 mol dm⁻³)

Method

1. Pipette 25.0 cm³ of the hydrochloric acid into the volumetric flask, make up to the mark with distilled water, stopper and invert to mix.
2. Pipette 25.0 cm³ of the diluted acid into a conical flask and add indicator.
3. Titrate with the sodium hydroxide from the burette to the end-point.
4. Repeat to obtain concordant titres and calculate the mean.

5. Calculate the concentration of the diluted acid, then multiply by the **dilution factor** (10) to get the concentration of the original acid.

Measurement Uncertainty

Apparatus	Typical uncertainty	% uncertainty example
Burette (per reading)	$\pm 0.05 \text{ cm}^3$	Two readings per titre: $(2 \times 0.05 / 24.00) \times 100 = 0.42\%$
25.0 cm^3 pipette	$\pm 0.06 \text{ cm}^3$	$(0.06 / 25.0) \times 100 = 0.24\%$
250 cm^3 volumetric flask	$\pm 0.3 \text{ cm}^3$	$(0.3 / 250) \times 100 = 0.12\%$

The total percentage uncertainty is found by **adding** the percentage uncertainties of each measurement. Percentage error compares your result with an accepted value: $(|\text{experimental} - \text{accepted}| / \text{accepted}) \times 100$. If the percentage error is larger than the total percentage uncertainty, there must be other (usually procedural) errors.

Safety

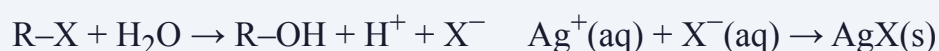
- Hydrochloric acid and sodium hydroxide are corrosive or irritant depending on concentration — wear eye protection.

Exam Tip: Do not forget the dilution factor in the final step. To reduce the percentage uncertainty in the titre, use a larger titre (e.g. more dilute NaOH or a larger sample). Know the difference between **measurement uncertainty** (the limit of the apparatus) and **measurement error** (difference from the true value).

CP4: Investigation of the Rates of Hydrolysis of Some Halogenoalkanes

Background Theory

Halogenoalkanes are hydrolysed by water to form alcohols and halide ions. Ethanol is used as a **common solvent** because halogenoalkanes do not mix with water. The halide ions released react with silver ions to give a precipitate, so the **time for a precipitate to appear** measures the rate of hydrolysis.



Two comparisons are made: (1) **different halogens** (1-chloro-, 1-bromo- and 1-iodobutane), and (2) **primary, secondary and tertiary** structures (1-bromobutane, 2-bromobutane and 2-bromo-2-methylpropane).

Apparatus

- Water bath at about 50 degrees C, test tubes with loose bungs, test-tube rack
- Ethanol, aqueous silver nitrate solution, the six halogenoalkanes, stop clock

Method

1. Put 5 cm³ of ethanol in each of three test tubes and add four drops of a different halogenoalkane to each. Loosely bung and stand them in the water bath.
2. Put 5 cm³ of silver nitrate solution in three more tubes and warm them in the same bath.
3. When all the tubes are at the bath temperature, add a tube of silver nitrate solution to one halogenoalkane solution and start the clock.
4. Stop the clock when the mixture first turns **cloudy**. Record the colour of the precipitate.
5. Repeat for the other two halogenoalkanes, then repeat the whole procedure with the primary, secondary and tertiary bromoalkanes.

Expected Results

Comparison	Order of rate (fastest first)	Explanation
Different halogens	Iodo > bromo > chloro	C–I bond enthalpy is lowest, so it breaks most easily. Bond enthalpy matters more than bond polarity here.
Different structures	Tertiary > secondary > primary	Tertiary halogenoalkanes react by an S _N 1 mechanism via a relatively stable tertiary carbocation
Precipitate colours	AgCl white, AgBr cream, AgI yellow	Confirms which halide has been released

Safety

- Halogenoalkanes and ethanol are **flammable** and harmful — no naked flames; use a water bath.
- Silver nitrate solution is corrosive or irritant and stains skin — wear eye protection and gloves.

Error Analysis

Source of Error	Type	Improvement
Deciding when the solution first becomes cloudy	Random	Place tubes against a dark background with a printed mark behind; same observer each time
Different temperatures for different tubes	Systematic	Pre-warm all solutions in the same bath and keep tubes in the bath during the reaction
Drops of different size	Random	Use a graduated pipette to add equal volumes
1-chlorobutane may not react in a reasonable time	—	Set a cut-off time and record as "no precipitate within x minutes"

Exam Tip: Water, not hydroxide ions, is the nucleophile here, so the reaction is slow enough to time. The rate depends on **C–X bond enthalpy**, not polarity: C–Cl is the most polar bond

but reacts slowest. Ethanol is a common solvent, so the halogenoalkane and the aqueous silver nitrate mix in one phase. Rate is proportional to $1/\text{time}$.

CP5: The Oxidation of Ethanol

Background Theory

Ethanol, a primary alcohol, is oxidised by **acidified potassium dichromate(VI)**. The orange $\text{Cr}_2\text{O}_7^{2-}$ ions are reduced to green Cr^{3+} ions. The product depends on the conditions:

- **Distil the product off as it forms** (alcohol in excess): **ethanal** (b.p. 21 degrees C) is removed before it can be oxidised further.
- **Heat under reflux** with excess oxidising agent, then distil: **ethanoic acid** (b.p. 118 degrees C) is formed.



Apparatus

- Pear-shaped or round-bottomed flask, Liebig condenser, still head, thermometer, receiver flask
- Anti-bumping granules, water bath or electric heater, ice bath
- Potassium dichromate(VI), dilute and concentrated sulfuric acid, ethanol

Method: Making Ethanoic Acid (Reflux Then Distil)

1. Put the acidified dichromate(VI) solution in the flask with anti-bumping granules, and cool in an ice bath.
2. Add ethanol a few drops at a time with swirling (the reaction is exothermic).
3. Fit a **vertical condenser** and heat under reflux for about 15–20 minutes. The vapour condenses and returns to the flask, so volatile materials are not lost and oxidation goes to completion.

4. Rearrange the apparatus for distillation and collect the liquid that distils at about 118 degrees C.
5. Test the distillate: it turns blue litmus red and fizzes with sodium carbonate or sodium hydrogencarbonate.

Method: Making Ethanal (Distil Immediately)

1. Set up for distillation with the receiver cooled in ice.
2. Warm the acidified dichromate(VI) and add ethanol dropwise, so the ethanal distils out as soon as it forms.
3. Test the distillate with Fehling's or Benedict's solution (blue to red precipitate) or Tollens' reagent (silver mirror).

Safety

- Potassium dichromate(VI) is **toxic, carcinogenic and oxidising** — wear gloves and eye protection.
- Concentrated sulfuric acid is corrosive; ethanol and ethanal are highly flammable — heat with a water bath or electric heater, never a naked flame.
- Never heat a closed system: the top of the reflux condenser must be open.

Error Analysis

Source of Error	Type	Improvement
Ethanal escapes as vapour	Systematic (low yield)	Cool the receiver in ice; keep joints sealed
Oxidation incomplete when ethanoic acid is wanted	Systematic (product contains ethanal/ethanol)	Use excess oxidising agent and reflux for longer
Distillate collected over too wide a temperature range	Systematic (impure product)	Collect only within a few degrees of the boiling temperature

Exam Tip: Be able to draw and label both reflux (vertical condenser, open top) and distillation (sloping condenser, thermometer bulb at the side-arm) apparatus. Explain the choice: distil immediately to stop at the aldehyde; reflux with excess oxidant to reach the carboxylic acid. Water enters the condenser at the **bottom**. The colour change orange → green shows the alcohol has been oxidised.

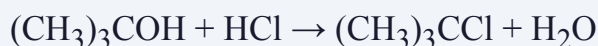
CP6: Chlorination of 2-Methylpropan-2-ol Using Concentrated Hydrochloric Acid

Background Theory

This core practical tests the techniques used to prepare and purify a liquid organic product. Many organic reactions require **prolonged heating** to go to completion, but the reagents and products are often volatile and would be lost if heated in an open flask. **Reflux** solves this: a vertical condenser is attached to the flask so that vapours condense and drip back into the reaction mixture, allowing continuous heating without loss of material. After the reaction is complete, the product is separated by **distillation**.

Method: 2-Chloro-2-methylpropane from 2-Methylpropan-2-ol

The tertiary alcohol reacts with concentrated hydrochloric acid at room temperature, so this preparation needs **no reflux**:



1. In a fume cupboard, put the 2-methylpropan-2-ol and an excess of concentrated hydrochloric acid in a **separating funnel**.
2. Stopper and shake for about 20 minutes, inverting and opening the tap frequently to **release pressure**.
3. Let the layers separate. The halogenoalkane is less dense than the aqueous layer, so it forms the **upper layer**. Run off and discard the lower aqueous layer.

4. Wash the organic layer with **sodium hydrogencarbonate solution** to remove remaining acid (vent often: CO_2 is produced), then with water. Discard the aqueous layer each time.
5. Run the organic layer into a dry flask and dry it with an **anhydrous salt** (e.g. anhydrous sodium sulfate) until the liquid is clear.
6. Decant into a clean flask and **distil**, collecting the fraction boiling at about 50–52 degrees C.

The more general techniques below (reflux, washing, drying, redistillation) are also examined in this core practical and in CP5.

General Technique: Reflux

1. Place reagents in a round-bottomed flask. Add anti-bumping granules.
2. Attach a **vertical condenser** (open at the top — not sealed) to the flask. Water flows in at the bottom and out at the top.
3. Heat the mixture gently (water bath for flammable liquids, or electric heating mantle). Vapours rise, condense in the condenser, and **return to the flask**. Continue for the required time (e.g. 30–60 minutes).
4. After reflux, rearrange the apparatus for **distillation**: tilt the condenser to an angle, connect to a collecting flask, and distil the product at its boiling point.

Purification Steps (After Distillation)

1. **Washing:** Use a separating funnel to wash the organic product with aqueous solutions to remove impurities. For example, wash with NaHCO_3 (removes acidic impurities) or dilute acid (removes basic impurities). Shake, vent, and drain off the aqueous layer.
2. **Drying:** Add an anhydrous drying agent (e.g. anhydrous MgSO_4 , CaCl_2 , or Na_2SO_4) to the organic layer. Leave for 10–15 minutes, swirling occasionally. The drying agent absorbs residual water. Filter off the drying agent by gravity filtration.
3. **Redistillation:** Distil the dried product, collecting the fraction at the correct boiling point (± 2 degrees C). This gives the purest product.

Purity Testing

- **Boiling point:** A pure liquid has a sharp, constant boiling point matching the literature value. Impurities change the boiling temperature and make the liquid boil over a range.

- **Refractive index:** Measured using a refractometer. Each pure liquid has a characteristic refractive index. Impurities alter the value.
- **GC (gas chromatography):** A pure substance shows a single peak. Impurities appear as additional peaks.

Safety

- Organic liquids are **flammable** — use electric heating (water bath or heating mantle). No naked flames.
- Never heat a sealed system — the condenser must be open at the top during reflux.
- When using a separating funnel: **vent regularly** by inverting and opening the tap to release gas pressure build-up.
- Some drying agents (anhydrous CaCl_2) react with alcohols and cannot be used for drying them.

Exam Tip: Reflux = heating without losing vapour (condenser vertical, vapour returns to flask). Distillation = separating by boiling point (condenser angled, vapour collected). The thermometer bulb should be at the **side-arm outlet** during distillation. When using a separating funnel, the organic layer may be on top (less dense than water) or bottom (if denser, e.g. DCM). Know which layer to keep! For drying: anhydrous Na_2SO_4 is a general-purpose drying agent suitable for most organic liquids.

CP7: Analysis of Some Inorganic and Organic Unknowns

Background Theory

This practical uses simple test-tube reactions to identify unknown substances: organic liquids by their **functional groups**, and inorganic solids by their **cations and anions**. It brings together tests met in Topics 4 and 6.

Tests on Organic Liquids

Test	Positive result	Shows
Shake with bromine water	Orange/brown → colourless	Alkene (C=C)
Warm with acidified potassium dichromate(VI) in a water bath	Orange → green	Primary or secondary alcohol (or aldehyde)
Heat with Fehling's (or Benedict's) solution in a water bath	Blue solution → red precipitate	Aldehyde
Warm with ethanol and NaOH(aq), acidify with dilute HNO ₃ , add AgNO ₃ (aq)	White / cream / yellow precipitate	Chloro- / bromo- / iodoalkane
Add sodium hydrogencarbonate or carbonate	Effervescence ; gas turns limewater milky	Carboxylic acid
Add phosphorus(V) chloride (fume cupboard)	Steamy fumes of HCl	–OH group (alcohol or carboxylic acid)

Tests on Inorganic Solids

Test	Positive result	Shows
Flame test (clean nichrome wire dipped in conc. HCl, then solid)	Li ⁺ red; Na ⁺ yellow; K ⁺ lilac; Ca ²⁺ orange-red; Sr ²⁺ red; Ba ²⁺ pale green	Group 1 or 2 cation
Warm with NaOH(aq); hold damp red litmus at the mouth	Litmus turns blue (ammonia)	NH ₄ ⁺
Add dilute acid; bubble gas through limewater	Effervescence; limewater turns milky	CO ₃ ²⁻ (or HCO ₃ ⁻)
Acidify with dilute HNO ₃ (or HCl), add BaCl ₂ (aq)	White precipitate of BaSO ₄	SO ₄ ²⁻

Acidify with dilute HNO_3 , add $\text{AgNO}_3(\text{aq})$, then dilute $\text{NH}_3(\text{aq})$	White ppt, soluble in dilute NH_3 / cream ppt, soluble in conc. NH_3 / yellow ppt, insoluble	Cl^- / Br^- / I^-
Add chlorine water (then shake with a hydrocarbon solvent)	Solution turns yellow-orange (Br_2) or brown (I_2)	Br^- or I^- (displacement)

Safety

- Treat all unknowns as harmful; treat unknown organic liquids as **flammable** — heat with a water bath.
- Potassium dichromate(VI) is carcinogenic and oxidising; chlorine water and bromine water are toxic — wear gloves and use a fume cupboard.
- Silver nitrate is corrosive and stains skin; barium chloride is harmful.

Exam Tip: Edexcel frequently asks you to identify an unknown from a set of test results. Approach systematically: flame test for the cation, then acidified AgNO_3 and BaCl_2 for the anion. Always explain why acid is added first: nitric acid removes carbonate ions, which would otherwise give a precipitate with Ag^+ or Ba^{2+} . Do not use HCl before AgNO_3 , because it adds chloride ions.

CP8: Determine the Enthalpy Change of a Reaction Using Hess's Law

Background Theory

Some enthalpy changes cannot be measured directly. The thermal decomposition of potassium hydrogencarbonate is an example: it needs continuous heating, so its temperature change cannot be measured with a simple calorimeter.



Instead, the enthalpy changes of two reactions that **can** be measured are found, and **Hess's law** is used to combine them. Both solids react quickly with excess dilute hydrochloric acid:

- Reaction 1: $\text{KHCO}_3(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{KCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ (ΔH_1 , endothermic: temperature falls)
- Reaction 2: $\text{K}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{KCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ (ΔH_2 , exothermic: temperature rises)

$$\Delta_{\text{r}}H = 2\Delta H_1 - \Delta H_2$$

Apparatus

- Expanded polystyrene cup in a beaker (for stability), lid, thermometer reading to 0.1 or 0.2 degrees C
- 50 cm³ measuring cylinder or burette, weighing bottles, balance
- About 2 mol dm⁻³ hydrochloric acid (in excess), potassium hydrogencarbonate, anhydrous potassium carbonate

Method

1. Put 30 cm³ of the acid in the polystyrene cup and record its temperature every minute for a few minutes until steady.
2. Weigh about 3 g of potassium hydrogencarbonate in a weighing bottle.
3. Add the solid to the acid at a recorded time, stir, and keep recording the temperature every half-minute or minute for several minutes.
4. Reweigh the empty weighing bottle to find the exact mass added.
5. Repeat with fresh acid and about 3 g of anhydrous potassium carbonate.
6. For each reaction, plot temperature against time and **extrapolate** the cooling (or warming) line back to the time of mixing to find the corrected ΔT .

Calculation

$q = m c \Delta T$ (m = mass of acid solution, taking density 1.00 g cm^{-3} ; $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

$\Delta H = -q / n$, where n = moles of the solid (the acid is in excess). Include the sign: positive for reaction 1 (temperature fell), negative for reaction 2.

Then $\Delta_r H = 2\Delta H_1 - \Delta H_2$ (the value is positive: the decomposition is endothermic).

Safety

- Hydrochloric acid at this concentration is an irritant — wear eye protection.
- Add the solids gradually if the cup is small, as the effervescence can cause the mixture to froth over.

Error Analysis

Source of Error	Type	Improvement
Heat exchange with the surroundings	Systematic	Lid and insulation; graphical extrapolation to the time of mixing
Acid spray lost with the escaping CO_2	Systematic	Add solid in portions; loose lid
Small ΔT with a thermometer reading to 0.5 degrees C	Random (large % uncertainty)	Use a 0.1 degrees C thermometer or temperature probe; use more solid
Assuming c and density equal those of water	Systematic	Acknowledge as a limitation

Exam Tip: Draw the Hess cycle with $2\text{KHCO}_3 + 2\text{HCl}$ and $\text{K}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2 + 2\text{HCl}$ at the top, and $2\text{KCl} + 2\text{H}_2\text{O} + 2\text{CO}_2$ at the bottom. Remember to **double** ΔH_1 , because the equation uses 2 mol of KHCO_3 . Give the sign and units (kJ mol^{-1}) in your final answer. The acid must be in excess so that all of the solid reacts.

CP9: Finding the K_a Value for a Weak Acid

Background Theory

For a weak acid $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$, $K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$. When exactly **half** of the acid has been neutralised by a strong base, $[\text{HA}] = [\text{A}^-]$, so these terms cancel and $K_a = [\text{H}^+]$. Therefore at half-neutralisation:

$$\text{pH} = \text{p}K_a \quad K_a = 10^{-\text{pH}}$$

In this practical, ethanoic acid is titrated with sodium hydroxide to find the volume needed for full neutralisation. An identical volume of acid is then half-neutralised, and the pH of the resulting buffer is measured.

Apparatus

- Calibrated pH meter (buffer solutions for calibration)
- 25.0 cm³ pipette and filler, 50 cm³ burette, conical flasks, beakers
- About 0.1 mol dm⁻³ ethanoic acid and sodium hydroxide, phenolphthalein indicator

Method

1. **Calibrate** the pH meter using buffer solutions of known pH.
2. Pipette 25.0 cm³ of ethanoic acid into a conical flask, add phenolphthalein, and titrate with NaOH to the first permanent pink colour. Repeat for concordant results; call the mean titre V.
3. Pipette another 25.0 cm³ of the acid into a beaker and add exactly **V/2** of NaOH from the burette (no indicator). Stir.
4. Measure the pH of this half-neutralised solution. Rinse the electrode with distilled water between readings.
5. Alternative method: add exactly V of NaOH to a 25.0 cm³ portion of acid (fully neutralised), then add a further 25.0 cm³ of the acid. This mixture also contains equal amounts of HA and A⁻.

6. Calculate $K_a = 10^{-\text{pH}}$.

Safety

- Sodium hydroxide is corrosive to eyes; ethanoic acid is an irritant — wear eye protection.
- pH electrodes are fragile; keep them in storage solution when not in use.

Error Analysis

Source of Error	Type	Improvement
pH meter not calibrated	Systematic	Calibrate with at least two buffers either side of the expected pH
Not exactly half-neutralised	Systematic	Deliver $V/2$ from a burette; use the "add another equal portion of acid" method
Temperature different from 25 degrees C	Systematic (K_a depends on temperature)	Record the temperature and quote K_a at that temperature
Reading taken before it has stabilised	Random	Stir and wait until the reading is steady

Exam Tip: Explain why $\text{pH} = \text{p}K_a$ at half-neutralisation: $[\text{HA}] = [\text{A}^-]$, so they cancel in the K_a expression. The same idea lets you read $\text{p}K_a$ from a weak acid–strong base titration curve at half the equivalence volume. For ethanoic acid, expect $\text{p}K_a$ of about 4.8 (K_a about $1.7 \times 10^{-5} \text{ mol dm}^{-3}$).

CP10: Investigating Some Electrochemical Cells

Background Theory

An **electrochemical cell** is made by joining two half-cells with a **salt bridge** and connecting the electrodes through a high-resistance voltmeter. Each half-cell contains an oxidised and a reduced form of an element: a metal dipping into a solution of its ions (e.g. Cu/Cu^{2+}), or a platinum electrode in a solution containing two ions of the same element (e.g. $\text{Fe}^{3+}/\text{Fe}^{2+}$). The measured cell e.m.f. is compared with the value calculated from standard electrode potentials.

$$E^\circ_{\text{cell}} = E^\circ(\text{right-hand, reduction}) - E^\circ(\text{left-hand, oxidation})$$

For example, $\text{Zn}(\text{s})|\text{Zn}^{2+}(\text{aq})||\text{Cu}^{2+}(\text{aq})|\text{Cu}(\text{s})$: $E^\circ_{\text{cell}} = +0.34 - (-0.76) = \mathbf{+1.10\text{ V}}$.

Apparatus

- Small beakers, strips of copper, zinc, iron and other metals, emery paper
- 1.0 mol dm^{-3} solutions of the metal ions (e.g. CuSO_4 , ZnSO_4); a solution containing both Fe^{2+} and Fe^{3+} with a platinum electrode
- Salt bridge: strip of filter paper soaked in saturated potassium nitrate solution
- High-resistance voltmeter, leads and crocodile clips

Method

1. Clean each metal strip with emery paper to remove the oxide layer.
2. Set up each half-cell by standing the electrode in about 50 cm^3 of the solution of its ions.
3. Connect two half-cells with a fresh salt bridge (made just before use) and connect the electrodes to the voltmeter.
4. Record the e.m.f. and note which electrode is the positive terminal.
5. Repeat for different combinations of half-cells, and compare each measured value with the value calculated from the data booklet.

Safety

- Copper(II) sulfate and zinc sulfate are harmful and irritant — wear eye protection and wash hands.
- Potassium nitrate is an oxidising agent; keep it away from flammables.
- Dispose of metal ion solutions as instructed, not down the sink.

Error Analysis

Source of Error	Type	Improvement
Oxide layer on metal surface	Systematic (e.m.f. lower than expected)	Clean with emery paper immediately before use
Concentrations not 1.0 mol dm^{-3} or temperature not 298 K	Systematic (non-standard conditions)	Make solutions accurately; record the temperature
Voltmeter draws current	Systematic (reading too low)	Use a high-resistance voltmeter
Salt bridge dries out or is contaminated	Random	Use a fresh salt bridge for each cell

Exam Tip: Potassium nitrate is used in the salt bridge because its ions do not react with the solutions or form precipitates (KCl would precipitate AgCl with a silver half-cell). A wire cannot replace the salt bridge because the circuit needs ions, not electrons, to move between the solutions. A positive E°_{cell} means the reaction is thermodynamically feasible, but it may still be too slow to observe (kinetic inhibition), and non-standard conditions change the actual value.

CP11: Redox Titration (Iron(II) with Manganate(VII))

Background Theory

Potassium manganate(VII) is a strong oxidising agent in acid solution. It oxidises iron(II) ions to iron(III) ions and is itself reduced to almost colourless Mn^{2+} . A typical use is to find the percentage of iron in iron tablets.



No indicator is needed: the manganate(VII) is **self-indicating**. The end-point is the first **permanent pale pink** colour, showing a slight excess of MnO_4^- .

Apparatus

- Iron tablets, mortar and pestle, balance, beaker, 250 cm³ volumetric flask
- 25.0 cm³ pipette and filler, 50 cm³ burette, conical flasks, white tile
- About 0.01 mol dm⁻³ potassium manganate(VII) solution, 1 mol dm⁻³ sulfuric acid

Method

1. Weigh several iron tablets, grind them, and dissolve the powder in dilute sulfuric acid (a little undissolved coating is normal).
2. Transfer the solution and all washings to a 250 cm³ volumetric flask and make up to the mark with dilute sulfuric acid or distilled water. Mix well.
3. Pipette 25.0 cm³ into a conical flask and add about 10 cm³ of dilute sulfuric acid.
4. Titrate with the manganate(VII) solution until the first permanent pale pink colour appears.
5. Repeat for concordant titres.

Calculation

Moles $\text{MnO}_4^- = c \times (\text{mean titre} / 1000)$

Moles Fe^{2+} in 25.0 cm³ = 5 x moles MnO_4^-

Moles Fe^{2+} in 250 cm³ = 10 x that value

Mass of iron = moles x 55.8; % iron = (mass of iron / mass of tablets) x 100

Safety

- Potassium manganate(VII) is oxidising and harmful and stains skin; sulfuric acid is an irritant — wear eye protection.

Error Analysis

Source of Error	Type	Improvement
Reading the burette: the dark purple solution hides the meniscus	Systematic	Read the top of the meniscus consistently
Fe ²⁺ oxidised by air before titration	Systematic (titre too low)	Titrate promptly after dissolving
Not enough acid added	Systematic (brown MnO ₂ forms)	Add excess dilute sulfuric acid to each flask

Exam Tip: Use dilute **sulfuric** acid only. Hydrochloric acid would be oxidised by manganate(VII) to chlorine, giving a titre that is too large; nitric acid is itself an oxidising agent and could oxidise the Fe²⁺. The mole ratio MnO₄⁻ : Fe²⁺ is **1 : 5**. The specification also expects iodine–thiosulfate titration calculations (I₂ + 2S₂O₃²⁻ → 2I⁻ + S₄O₆²⁻, with starch added near the end-point).

CP12: The Preparation of a Transition Metal Complex

Background Theory

Adding concentrated ammonia to copper(II) sulfate solution causes **ligand exchange**: four water ligands are replaced by ammonia, giving the deep blue tetraamminecopper(II) complex. The solid complex, **tetraamminecopper(II) sulfate-1-water**, is much less soluble in ethanol than in water, so adding ethanol makes it crystallise out.



Apparatus

- Test tube, balance, hot water bath, graduated pipettes, small beaker, ice bath
- Buchner funnel and flask (vacuum filtration), filter papers
- Copper(II) sulfate-5-water, concentrated ammonia solution, ethanol

Method

1. Weigh about 1.5 g of copper(II) sulfate-5-water into a test tube (weigh by difference).
2. Add about 4 cm³ of water and warm in a hot water bath, stirring, to dissolve.
3. In a **fume cupboard**, add about 2 cm³ of concentrated ammonia solution while stirring. A pale blue precipitate forms first, then dissolves to give a deep blue solution.
4. Pour the solution into about 6 cm³ of ethanol in a beaker, mix, and cool in an ice bath until dark blue crystals form.
5. Collect the crystals by vacuum filtration. Rinse the tube with cold ethanol onto the funnel, then wash the crystals with a little cold ethanol.
6. Dry the crystals between filter papers and weigh them.

Calculation

$$\text{Moles CuSO}_4 \cdot 5\text{H}_2\text{O} = \text{mass} / 249.7$$

$$\text{Theoretical mass of product} = \text{moles} \times 245.8 \text{ (M of } [\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot \text{H}_2\text{O)}$$

$$\% \text{ yield} = (\text{actual mass} / \text{theoretical mass}) \times 100$$

Safety

- Concentrated ammonia is **corrosive** and gives off a toxic, choking gas — use a fume cupboard, gloves and eye protection.
- Ethanol is highly flammable — no naked flames.
- Copper(II) sulfate is harmful — avoid skin contact.

Error Analysis

Source of Error	Type	Improvement
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Product remains dissolved	Systematic (low yield)	Cool thoroughly in ice; use cold ethanol for washing
Crystals still damp when weighed	Systematic (yield too high)	Dry to constant mass
Crystals lost during transfer	Systematic (low yield)	Rinse the tube with cold ethanol onto the filter

Exam Tip: The pale blue precipitate that forms first is copper(II) hydroxide (ammonia acting as a base). Excess ammonia then causes ligand exchange, giving the deep blue $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ ion in solution. Ethanol is used because the complex is much less soluble in it than in water. Cold ethanol is used for washing so that little product redissolves. Yields above 100% usually mean the product is still wet.

CP13a: Rates of Reaction — Following the Iodine–Propanone Reaction by a Titrimetric Method

Background Theory

Propanone reacts with iodine in acid solution. The acid is a **catalyst**:



Samples are removed at intervals, the reaction in each sample is stopped (**quenched**), and the iodine remaining is found by titration with sodium thiosulfate. Propanone and acid are in large excess, so only $[\text{I}_2]$ changes significantly. The shape of the $[\text{I}_2]$ –time graph gives the order with respect to iodine.

Apparatus

- Beaker, stop clock, 10 cm³ pipette and filler, conical flasks, burette
- 1 mol dm⁻³ propanone, 1 mol dm⁻³ sulfuric acid, 0.02 mol dm⁻³ iodine solution
- Sodium hydrogencarbonate (to quench), 0.01 mol dm⁻³ sodium thiosulfate, starch indicator

Method

1. Mix 25 cm³ of propanone solution with 25 cm³ of sulfuric acid in a beaker.
2. Add 50 cm³ of iodine solution, **starting the clock** at the moment of mixing, and swirl.
3. Pipette out a 10 cm³ sample into a conical flask and add a spatula of sodium hydrogencarbonate. Record the exact time of quenching (leave the clock running).
4. Titrate the iodine in the sample with sodium thiosulfate, adding starch near the end-point (blue-black → colourless).
5. Take further samples about every 3 minutes and treat each in the same way.

Analysis

- The titre is proportional to [I₂] remaining. Plot titre (or [I₂]) against time.
- A **straight line** with constant negative gradient shows the rate does not depend on [I₂], so the reaction is **zero order** with respect to iodine.
- Further experiments varying [propanone] and [H⁺] show the reaction is first order in each: rate = k[CH₃COCH₃][H⁺]. Iodine is therefore not involved in the rate-determining step.

Safety

- Propanone is highly flammable; iodine solution is harmful and stains; sulfuric acid is an irritant — wear eye protection, no naked flames.
- Iodopropanone formed is a lachrymator (irritates eyes) — work in a well-ventilated area.

Error Analysis

Source of Error	Type	Improvement
Time recorded at the wrong moment	Systematic	Record the time when the quenching solid is added, not when the sample is drawn

Reaction continues while sampling	Random	Quench immediately; practise the pipetting step
Starch added too early	Systematic (end-point unclear)	Add starch only when the solution is pale yellow

Exam Tip: Sodium hydrogencarbonate quenches the reaction by **neutralising the acid catalyst**. A straight-line concentration–time graph means zero order; a curve with constant half-life means first order. Because iodine is zero order, it must react **after** the rate-determining step. Be ready to propose a mechanism consistent with $\text{rate} = k[\text{propanone}][\text{H}^+]$.

CP13b: Rates of Reaction — Investigating a Clock Reaction (Iodine Clock)

Background Theory

In the **Harcourt–Esson** iodine clock, hydrogen peroxide oxidises iodide ions in acid solution:



A small, fixed amount of sodium thiosulfate is included. It immediately turns any iodine formed back into iodide ($\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$). When all the thiosulfate is used up, free iodine appears and reacts with starch to give a sudden **blue-black** colour. The time t is the time for a fixed amount of reaction, so the initial rate is proportional to **$1/t$** .

Apparatus

- Beakers or boiling tubes, burettes or graduated pipettes for each solution, stop clock, white tile

- Solutions of potassium iodide, sulfuric acid, sodium thiosulfate, starch and hydrogen peroxide; distilled water

Method

- In a beaker, mix measured volumes of sulfuric acid, starch, sodium thiosulfate, potassium iodide and water.
- Add a measured volume of hydrogen peroxide, start the clock and stir.
- Stop the clock when the blue-black colour appears.
- Repeat, changing the volume of **one** reactant (e.g. KI) and adding water so the **total volume stays the same**. Keep all other volumes and the temperature constant.
- Repeat for each reactant in turn to find the order with respect to each.

Determining Order of Reaction

Graph of $1/t$ against concentration (or volume) of X	Order with respect to X
Horizontal line	Zero order
Straight line through the origin	First order
Upward curve; straight line when $1/t$ is plotted against $[X]^2$	Second order

Alternatively, plot $\log(1/t)$ against $\log[X]$: the gradient equals the order.

Safety

- Hydrogen peroxide and sulfuric acid are irritants at these concentrations — wear eye protection.

Error Analysis

Source of Error	Type	Improvement
Judging the moment of colour change	Random	Same observer; white tile; or use a colorimeter

Temperature varies between runs	Systematic	Use a water bath and record the temperature
Total volume not kept constant	Systematic (all concentrations change)	Make up the difference with distilled water every time

Exam Tip: The thiosulfate must be the **limiting** reagent in small amount, so the colour appears after only a small fraction of the reaction: $1/t$ then approximates the **initial rate**. Doubling $[X]$ and doubling the rate means first order; quadrupling means second order; no change means zero order. The rate constant's units depend on the overall order.

CP14: Finding the Activation Energy of a Reaction

Background Theory

The **Arrhenius equation** links the rate constant to temperature:

$$k = Ae^{-E_a/RT} \rightarrow \ln k = -E_a/(RT) + \ln A$$

Pearson's version of this practical uses the reaction between **bromide ions and bromate(V) ions** in acid solution:



A small, fixed amount of **phenol** is added. It removes the bromine as fast as it forms:



When all the phenol has reacted, the bromine that is still being produced **bleaches the methyl red indicator**, so its red colour disappears. The time t is the time for the same fixed amount of

reaction at every temperature, so k is proportional to $1/t$. Substituting $k = c/t$ into the Arrhenius equation and rearranging gives:

$$\ln t = E_a/(RT) + \text{constant}$$

So a graph of **$\ln t$ against $1/T$** is a straight line with gradient **$+E_a/R$** .

Apparatus

- 0.01 mol dm^{-3} aqueous phenol solution
- Bromide/bromate(V) solution (0.1 mol dm^{-3} potassium bromide and 0.02 mol dm^{-3} potassium bromate(V) in the same solution)
- 0.5 mol dm^{-3} sulfuric acid, methyl red indicator
- Two boiling tubes, 10 cm^3 pipette, 500 cm^3 beaker as a water bath, kettle, ice, thermometer ($0\text{--}110$ degrees C), stop clock

Method

1. Pipette 10 cm^3 of phenol solution and 10 cm^3 of bromide/bromate(V) solution into one boiling tube, and add four drops of methyl red.
2. Pipette 5 cm^3 of sulfuric acid into a second boiling tube.
3. Use kettle water in the beaker to make a water bath at 75 degrees C (± 1 degrees C) and stand both tubes in it.
4. When both tubes have reached the bath temperature, mix them by pouring one into the other and back again, and start the stop clock at the same time.
5. Keep the reaction tube in the bath and stop the clock when the red colour of the methyl red disappears.
6. Repeat the whole experiment at 65 , 55 , 45 , 35 , 25 and 15 degrees C, using ice for the lowest temperature.

Analysis

- Record temperature and time, then add columns for T in kelvin (degrees C + 273), $1/T$ in K^{-1} , and $\ln t$.

- Plot **$\ln t$** (y-axis) against **$1/T$** (x-axis) and draw a line of best fit.
- Gradient = E_a/R , so **$E_a = \text{gradient} \times 8.31 \text{ J mol}^{-1}$** . Divide by 1000 for kJ mol^{-1} .
- If you plot $\ln(1/t)$ (which is proportional to $\ln k$) instead, the gradient is $-E_a/R$ and $E_a = -\text{gradient} \times R$. Both give the same value.

Safety

- Wear a lab coat, gloves and eye protection.
- Phenol is **toxic and corrosive**; avoid skin contact.
- Potassium bromate(V) is **oxidising**; sulfuric acid is an irritant.
- Take care with hot water from the kettle and the 75 degrees C bath.

Error Analysis

Source of Error	Type	Improvement
Temperature of the mixture changes during the run	Systematic	Keep the tube in the bath; record the temperature at the start and end and use the mean
Solutions not at bath temperature before mixing	Systematic	Leave both tubes in the bath for several minutes and check with the thermometer
Judging when the red colour disappears	Random	Same observer; white background; repeat and average
Very short times at high temperature (a few seconds)	Random (large % uncertainty in t)	Repeat these runs; give less weight to them when drawing the line

Exam Tip: Know the role of each reagent: phenol removes bromine so that nothing is seen until a fixed amount of reaction has happened, and methyl red shows when that point is reached by being bleached. Watch the sign of the gradient: $\ln t$ against $1/T$ has a **positive** gradient (E_a/R), while $\ln k$ or $\ln(1/t)$ against $1/T$ has a **negative** gradient ($-E_a/R$). E_a is always

positive. With $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$, E_a comes out in J mol^{-1} . The same analysis can be applied to other clock reactions, such as sodium thiosulfate with hydrochloric acid.

CP15: Analysis of Some Inorganic and Organic Unknowns

Background Theory

CP15 extends CP7. You plan and carry out tests to identify unknown substances, now including the A-level organic functional groups (carbonyl compounds and aldehydes) alongside the ion tests. Planning safe procedures, including hazard-based control measures, is part of the assessment.

Inorganic Ion Tests

Test	Positive result	Shows
Flame test	Characteristic colour (e.g. Na^+ yellow, K^+ lilac, Ca^{2+} orange-red)	Group 1/2 cation
Add NaOH(aq) dropwise, then in excess	Cu^{2+} blue ppt; Fe^{2+} green ppt (browns in air); Fe^{3+} brown ppt; Cr^{3+} grey-green ppt dissolving in excess; Al^{3+} white ppt dissolving in excess	Metal cation
Acidified $\text{AgNO}_3(\text{aq})$	White / cream / yellow precipitate	Cl^- / Br^- / I^-
Acidified $\text{BaCl}_2(\text{aq})$	White precipitate	SO_4^{2-}
Dilute acid, gas into limewater	Effervescence; limewater milky	CO_3^{2-}

Organic Functional Group Tests

Functional Group	Test	Positive Result
Alkene (C=C)	Shake with bromine water	Orange/brown → colourless
Carbonyl (aldehyde or ketone)	Add 2,4-dinitrophenylhydrazine (Brady's reagent)	Orange/yellow precipitate
Aldehyde (not ketone)	Warm with Tollens' reagent; or heat with Fehling's/Benedict's solution	Silver mirror; or blue → red precipitate
Primary/secondary alcohol, aldehyde	Warm with acidified $K_2Cr_2O_7$	Orange → green
Carboxylic acid	Add $NaHCO_3$ or Na_2CO_3	Effervescence (CO_2)
Halogenoalkane (C-X)	Warm with $NaOH(aq)$ and ethanol, acidify with HNO_3 , add $AgNO_3$	White (Cl), cream (Br), yellow (I) precipitate
CH_3CO- or $CH_3CH(OH)-$ group	Warm with iodine in alkali (iodoform test)	Pale yellow precipitate of CHI_3 (antiseptic smell)

Safety

- Read the hazard labels of the unknowns; heat anything flammable only in a water bath.
- 2,4-dinitrophenylhydrazine is harmful and flammable (solid is explosive when dry) — use the supplied solution with gloves.
- Tollens' reagent must be made fresh and washed away after use (it can form explosive compounds on standing).

Exam Tip: Order your tests logically: 2,4-DNPH first to show a carbonyl group, then Tollens' or Fehling's to decide aldehyde or ketone. The melting temperature of the purified 2,4-DNPH derivative can be compared with data to identify the specific carbonyl

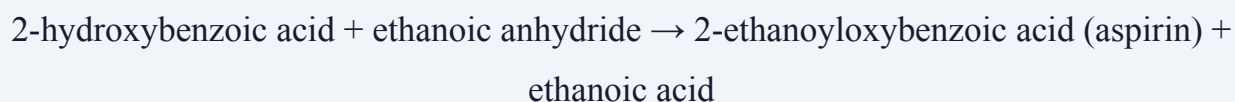
compound. Always describe **both** the reagent and conditions and the observation for a positive result.

CP16: The Preparation of Aspirin

Background Theory

Organic solids (here aspirin, an ester of 2-hydroxybenzoic acid) are prepared by organic synthesis reactions, then purified by **recrystallisation** and their purity assessed by **melting point determination**. A pure substance has a **sharp melting point** (melts over a narrow range, typically <2 degrees C). An impure substance melts over a wider range and at a lower temperature than expected (melting point depression).

Method: Preparing Aspirin



1. Weigh about 2 g of 2-hydroxybenzoic acid into a pear-shaped flask. In a fume cupboard, add about 4 cm³ of ethanoic anhydride and a few drops of concentrated phosphoric(V) acid (catalyst).
2. Warm the mixture in a hot water bath (about 85 degrees C) for about 5–10 minutes, swirling, to complete the reaction.
3. Carefully add a little cold water to destroy the excess ethanoic anhydride, then cool the flask in iced water until crystals form.
4. Collect the crude aspirin by vacuum filtration and wash with a little ice-cold water.
5. Recrystallise from the minimum volume of hot solvent (a mixture of ethanol and water works well), then dry and weigh.
6. Check purity by melting temperature: pure aspirin melts at about 136 degrees C.

Recrystallisation Method

1. **Dissolve** the crude product in the **minimum volume** of hot solvent (just enough to dissolve it at the boiling point). Common solvents: water, ethanol, ethyl ethanoate. The choice of solvent is critical: the product must be soluble in the hot solvent but insoluble in the cold solvent.
2. **Hot filter** (if necessary): filter the hot solution through a pre-warmed funnel with fluted filter paper. This removes insoluble impurities. Work quickly to prevent premature crystallisation.
3. **Cool slowly**: Allow the filtrate to cool slowly to room temperature, then place in an ice bath. Slow cooling produces larger, purer crystals (rapid cooling gives small, impure crystals because impurities get trapped).
4. **Filter under reduced pressure** (vacuum filtration using a Buchner funnel and flask). This is faster than gravity filtration and removes more solvent. The crystals are collected on the filter paper; the impurities remain dissolved in the cold filtrate (mother liquor).
5. **Wash** the crystals on the filter with a small volume of ice-cold solvent (to remove residual impurities from the crystal surfaces without re-dissolving the product).
6. **Dry** the crystals: press between sheets of filter paper, then leave in a desiccator (containing a drying agent like silica gel) or a warm oven (below the melting point).

Melting Point Determination

1. Pack a small amount of dry crystals into a **melting point capillary tube** (sealed at one end). The column of solid should be ~5 mm high.
2. Place the tube in an electrically heated **melting point apparatus** or attach it to a thermometer in a heated oil bath.
3. Heat slowly (~1–2 degrees C per minute near the expected melting point). Record the temperature at which the solid **first starts to melt** (meniscus appears) and the temperature at which it is **completely liquid**.
4. A **pure substance** melts over a narrow range (≤ 2 degrees C) and matches the literature value. An **impure substance** melts at a lower temperature and over a wider range.

Safety

- Hot solvents — risk of burns. Use a water bath to heat flammable solvents (never a naked flame).
- Organic solvents (ethanol, ethyl ethanoate) are flammable and produce vapours — good ventilation.

- Some organic products are irritants — wear gloves and goggles.
- Ethanoic anhydride and concentrated phosphoric(V) acid are corrosive; ethanoic anhydride vapour is irritating — use a fume cupboard.

Calculating Percentage Yield and Atom Economy

Percentage yield = (actual moles of product / theoretical moles of product) x 100

Reasons yield < 100%: incomplete reaction, side reactions, loss during transfer/filtration/washing, product left in mother liquor.

Atom economy = (M_r of desired product / sum of M_r of all products) x 100

A measure of how efficiently atoms are used. Higher atom economy = less waste.

Error Analysis

Source of Error	Type	Improvement
Product lost during recrystallisation (stays in solution)	Systematic (lowers yield)	Use the minimum volume of solvent; cool fully to maximise crystal recovery
Impurities not fully removed	Systematic (melting point too low and broad)	Repeat the recrystallisation; use a better solvent choice
Product not fully dry	Systematic (mass too high; melting point too low)	Dry thoroughly in desiccator; check mass constancy by re-weighing
Heating too fast during melting point	Systematic (overshoot; temperature reads too high)	Heat at 1–2 degrees C per minute near the expected melting point

Exam Tip: In recrystallisation: **minimum volume** of hot solvent ensures maximum yield. **Slow cooling** gives larger, purer crystals. **Vacuum filtration** (Buchner) is faster and gives

drier crystals than gravity filtration. A melting point range of >2 degrees C indicates impurity. Always compare your melting point with the literature value.

Supporting Content (Not Edexcel Core Practicals)

The sections below are not among the 16 core practicals in the Edexcel 9CH0 specification. They are kept because they support specification content: calculations using $pV = nRT$, calorimetry, rates from continuous monitoring, and pH curves.

Supporting Content: Molar Mass of a Volatile Liquid Using $pV = nRT$

Background Theory

The molar mass (M_r) of a volatile liquid can be determined experimentally using the **ideal gas equation: $pV = nRT$** . A known mass of the volatile liquid is vaporised at a known temperature and pressure, and the volume of the resulting gas is measured. Since $n = \text{mass}/M_r$, the equation can be rearranged to $M_r = mRT/(pV)$. Two methods are commonly used: the **gas syringe method** (inject liquid into a heated gas syringe) and the **Dumas method** (heat liquid in a flask with a pinhole until it all vaporises, then weigh the condensed vapour).

$$pV = nRT \rightarrow n = pV/RT \rightarrow M_r = mRT/(pV)$$

Where: p = pressure (Pa), V = volume (m^3), n = moles, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, T = temperature (K), m = mass (g)

Method (Gas Syringe)

1. Place a gas syringe (with a self-sealing rubber septum at one end) inside an oven set to a known temperature well above the boiling point of the liquid. Allow to reach thermal equilibrium. Record the temperature (T) in Kelvin and atmospheric pressure (p) in Pa.

2. Weigh the hypodermic syringe containing a small volume ($\sim 0.2 \text{ cm}^3$) of the volatile liquid. Record mass to $\pm 0.001 \text{ g}$.
3. Inject the liquid through the septum into the hot gas syringe. The liquid vaporises immediately. Re-weigh the hypodermic syringe.
4. Mass of liquid injected = initial mass – final mass.
5. Once the gas syringe has reached thermal equilibrium, record the volume of gas (V) in cm^3 . Convert to m^3 ($1 \text{ cm}^3 = 1 \times 10^{-6} \text{ m}^3$).
6. Calculate: $n = pV/(RT)$. Then $M_r = \text{mass}/n$.

Method (Dumas)

1. Place a few drops of the volatile liquid in a round-bottomed flask. Cover the opening with aluminium foil pierced with a **single pin-hole**. The pinhole allows vapour to escape but prevents air from entering rapidly.
2. Place the flask in a **boiling water bath** (100 degrees C / 373 K). As the liquid vaporises, excess vapour escapes through the pinhole until only vapour at atmospheric pressure remains in the flask.
3. When no more vapour is seen escaping, remove the flask and allow it to cool. The vapour condenses back to liquid.
4. Weigh the flask + condensed liquid + foil. Then weigh the empty, dry, clean flask + foil. Mass of vapour = difference.
5. Measure the volume of the flask by filling with water and measuring the volume of water. This gives V .
6. T = temperature of the water bath (in K); p = atmospheric pressure (in Pa).
7. Calculate $M_r = mRT/(pV)$.

Safety

- Volatile liquids are **highly flammable** — no naked flames; use electric oven/water bath.
- Handle hypodermic syringes with care (sharp needle — risk of needlestick injury).
- Oven is hot — use heat-resistant gloves.
- Work in a well-ventilated area (volatile vapours can be irritating).

Unit Conversions (Essential)

Quantity	Given In	Convert To	Conversion
Volume	cm ³	m ³	Multiply by 10 ⁻⁶
Temperature	degrees C	K	Add 273
Pressure	atm or kPa	Pa	1 atm = 101,325 Pa; 1 kPa = 1000 Pa

Error Analysis

Source of Error	Type	Improvement
Gas syringe friction (reads low volume)	Systematic (M_r too high)	Tap syringe gently to ensure free movement; use well-lubricated syringe
Temperature not uniform in oven	Systematic	Allow plenty of time for thermal equilibrium; place thermometer near the syringe
Liquid not fully vaporised	Systematic (volume too low, M_r too high)	Ensure oven temperature is well above the boiling point of the liquid
Vapour escaping around septum	Systematic (volume too low)	Check the septum seal is airtight before injecting
Ideal gas assumption	Systematic	Real gases deviate from ideality, especially at high pressures and low temperatures. The assumption is more valid at high T and low P. Acknowledge this limitation
Dumas: vapour escaping after removal from water bath	Systematic (mass too low)	Quickly seal the pinhole as soon as the flask is removed; cool rapidly

Exam Tip: Always convert **all units** before substituting into $pV = nRT$: volume to m^3 , temperature to K, pressure to Pa. The ideal gas equation assumes no intermolecular forces and that the gas molecules have negligible volume — these assumptions break down at high pressures and low temperatures. Experimental M_r values are typically slightly higher than true values because (1) some vapour may condense before the volume is read, and (2) gas syringe friction causes the volume to read low.

Supporting Content: Enthalpy of Neutralisation (Calorimetry)

Background Theory

A known volume of acid is mixed with a known volume of alkali in an insulated polystyrene cup, and the temperature change is used to calculate the enthalpy change per mole of water formed. The same calorimetry skills are used in CP8. Key points:

- The standard enthalpy of neutralisation for strong acid + strong base (e.g. $\text{HCl} + \text{NaOH}$) is approximately **$-57.1 \text{ kJ mol}^{-1}$** because the net ionic equation is the same: $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
- For reactions involving weak acids or bases, the measured ΔH is less exothermic because some energy is used to fully dissociate the weak acid/base (bond breaking is endothermic)
- Always use the **graphical extrapolation method** to find the corrected ΔT : plot temperature vs time, extrapolate the pre-mixing and post-mixing lines back to the time of mixing

$$q = mc\Delta T \quad \Delta H = -q/n \quad (\text{kJ mol}^{-1})$$

Exam Tip: Experimental values are always **less exothermic** than literature values due to heat loss. The two key assumptions: $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$ and density = 1.0 g cm^{-3} . For

neutralisation of a weak acid, ΔH is less than 57.1 kJ mol^{-1} because energy is needed to fully ionise the weak acid first.

Supporting Content: Enthalpy of Combustion of an Alcohol

Background Theory

The **enthalpy of combustion ($\Delta_c H$)** is the enthalpy change when one mole of a substance is completely burned in excess oxygen under standard conditions. This practical uses a simple **spirit burner calorimeter**: the alcohol burns and heats a known mass of water above it. By measuring the temperature rise of the water and the mass of alcohol burned, $\Delta_c H$ can be calculated. Experimental values are always significantly less exothermic than data book values, primarily due to substantial heat loss to the surroundings. Across a homologous series of alcohols, $\Delta_c H$ increases approximately linearly with each additional CH_2 group ($\sim 650 \text{ kJ mol}^{-1}$ per CH_2), because each additional CH_2 provides additional C–H and C–C bonds to be broken and additional C=O and O–H bonds to be formed.

Apparatus

- Spirit burner containing the alcohol to be tested (e.g. methanol, ethanol, propan-1-ol, butan-1-ol)
- Copper calorimeter (or metal can — copper has high thermal conductivity, transferring heat efficiently to water)
- Thermometer (± 0.5 degrees C) or temperature probe
- Measuring cylinder (100 cm^3), balance ($\pm 0.01 \text{ g}$)
- Draught shield (to reduce heat loss from wind), clamp stand, stirring rod

Method

1. Measure **100 cm^3** of cold water into the copper calorimeter. Record the **initial temperature** (T_1).

2. Weigh the spirit burner (with cap on) and record the mass to ± 0.01 g (m_1).
3. Position the calorimeter above the spirit burner. Surround with the draught shield.
4. Light the spirit burner. Stir the water gently with the stirring rod throughout.
5. When the temperature has risen by approximately **20 degrees C**, extinguish the flame by replacing the cap. Record the **maximum temperature** (T_2).
6. Immediately re-weigh the spirit burner (m_2).
7. Calculate:
 - $\Delta T = T_2 - T_1$
 - Mass of fuel burned = $m_1 - m_2$
 - $q = mc\Delta T$ ($m = 100$ g water, $c = 4.18$ J g⁻¹ K⁻¹)
 - Moles of fuel = $(m_1 - m_2) / M_r$
 - $\Delta_c H = -q / n$ (kJ mol⁻¹)
8. Repeat for different alcohols in the same homologous series.

Safety

- Alcohols are **highly flammable** — keep containers away from the flame; cap the burner promptly.
- Spirit burner flames can be **almost invisible** (especially methanol) — handle carefully; assume the burner is lit until confirmed otherwise.
- Hot apparatus — use tongs and a heatproof mat.

Analysis

- Plot **$\Delta_c H$ (y-axis)** against **number of carbon atoms (x-axis)**. Expect a linear trend: each additional CH₂ group adds ~650 kJ mol⁻¹ to the combustion enthalpy.
- Compare experimental values with data book values. Experimental values will be **much less exothermic** (typically 50–70% of literature values) due to the dominant source of error: heat loss.

Error Analysis

Source of Error	Type	Effect on $\Delta_c H$	Improvement
Heat loss to surroundings (dominant error)	Systematic	ΔT too small; $\Delta_c H$ less exothermic	Better insulation; draught shield; use a bomb calorimeter (industrial standard)
Incomplete combustion (soot/CO produced)	Systematic	Less heat released per mole; $\Delta_c H$ less exothermic	Ensure adequate air supply; trim and clean the wick; observe flame colour (yellow/sooty = incomplete)
Evaporation of alcohol from hot wick after extinguishing	Systematic	Mass loss overstated; n too high; $\Delta_c H$ appears less exothermic	Cap the burner promptly; re-weigh quickly
Heat absorbed by calorimeter, stirrer, and thermometer	Systematic	ΔT of water understated	Use a copper calorimeter (low heat capacity); determine a calorimeter constant
Water evaporation during heating	Systematic	Mass of water decreases; less water to heat	Use a lid on the calorimeter to reduce evaporation

Exam Tip: Experimental $\Delta_c H$ values are always **less exothermic** (smaller magnitude) than data book values. The main reason is **heat loss to the surroundings**. Incomplete combustion also contributes (producing CO and C instead of only CO₂, releasing less energy). The **linear trend** across the homologous series shows that each additional CH₂ group contributes approximately the same amount of energy. Use bond enthalpy calculations to explain the theoretical trend: breaking C–H and C–C bonds (endothermic) and forming C=O and O–H bonds (exothermic).

Supporting Content: Rate by Continuous Monitoring (Gas Collection)

The volume of gas produced (collected in a gas syringe) or the mass lost from an open flask on a balance is recorded at regular intervals. Key points: the initial rate = gradient of the tangent at $t = 0$. Compare curves for different concentrations, surface areas, or temperatures. The curve levels off when one reactant is consumed.

Exam Tip: On a volume-time graph: steeper initial gradient = faster rate. If the **same moles of limiting reagent** are used, all curves reach the **same final volume**. Increasing concentration makes the reaction faster (steeper curve) but does not change the total amount of product if the limiting reagent amount is unchanged.

Supporting Content: pH Curves for Acid-Base Titrations

A pH curve is obtained by adding a titrant from a burette in small portions and recording the pH with a calibrated pH meter after each addition. The weak acid–strong base curve is used in CP9 to find K_a . Key points:

Four Titration Curve Types

Titration	Initial pH	Equivalence pH	Indicator	Curve Shape
Strong acid + strong base	~1	7	Methyl orange or phenolphthalein	Steep vertical section at equivalence
Strong acid + weak base	~1	<7	Methyl orange only	Steep but shorter vertical section

Weak acid + strong base	~3	>7	Phenolphthalein only	Buffer region; half-equivalence: $\text{pH} = \text{pK}_a$
Weak acid + weak base	~3	~7	No suitable indicator (use pH meter)	Very gradual change; no distinct vertical section

Key Concepts

- **Buffer region:** The flat section before equivalence in a weak acid/strong base titration. The mixture of HA and A^- resists pH changes.
- **Half-equivalence point:** $[\text{HA}] = [\text{A}^-]$, so $\text{pH} = \text{pK}_a$. Read directly from the graph.
- **Equivalence point:** All the acid has reacted. For weak acid/strong base, $\text{pH} > 7$ (conjugate base hydrolysis). For strong acid/weak base, $\text{pH} < 7$.
- **Indicator choice:** The indicator's colour-change range must fall within the steep vertical section of the curve. If it does not, the end-point will not coincide with the equivalence point.

Exam Tip: You must be able to **sketch all four titration curves** and label key features (initial pH, buffer region, half-equivalence, equivalence point, final pH). At the half-equivalence point for a weak acid: **$\text{pH} = \text{pK}_a$** . This is the easiest experimental method to determine pK_a . Henderson-Hasselbalch equation: $\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$.